

Certificate in Accounting and Finance Stage Examination

Management Accounting

Model paper

100 marks



Instructions to examinees:

- Answer all **NINE** questions.
- The overall duration of the exam is 3 hours & 15 minutes, which includes the 15-minute reading time.
- Answer in **black** pen only.
- Answer Multiple-Choice Questions in the answer script only.

SECTION A (Ranging from 45 to 55 marks)

QUESTION 1

Select the most appropriate answer from the options provided for each of the following Multiple-Choice Questions (MCQs).

- (i) In a cost ledger system, the following data is available:

- Raw materials purchased: Rs. 100,000
- Indirect materials issued: Rs. 15,000
- Direct materials issued: Rs. 85,000
- Direct labour: Rs. 60,000
- Production overhead absorbed: Rs. 75,000

What will be the total debit to the Work-in-Process (WIP) Control Account?

- (a) Rs. 145,000 (b) Rs. 220,000 (c) Rs. 235,000 (d) Rs. 250,000

(01 mark)

- (ii) A job cost sheet shows the following data:

- Direct materials: Rs. 90,000
- Direct labour: 6,000 hours at Rs. 10 per hour = Rs. 60,000
- Overheads applied at Rs. 15 per direct labour hour

On review, it was discovered that 1,000 labour hours were idle time. The idle time wages had been included in direct labour, but they should have been treated as part of overheads.

What is the corrected total cost of the job?

- (a) Rs. 240,000 (b) Rs. 230,000 (c) Rs. 225,000 (d) Rs. 220,000

(02 marks)

- (iii) Which of the following journal entries correctly records the transfer of completed job to Finished Goods, where the cost is Rs. 250,000?

	Debit	Credit
(a) Finished Goods A/c To Work-in-Process A/c	250,000	250,000
(b) Work-in-Process A/c To Finished Goods A/c	250,000	250,000
(c) Cost of Sales A/c To Finished Goods A/c	250,000	250,000
(d) Cost of Sales A/c To Work-in-Process A/c	250,000	250,000

(01 mark)

- (iv) Which of the following situations would invalidate the assumptions of the EOQ model?
- Demand is steady and predictable throughout the year
 - Lead time is known and constant
 - Supplier offers a bulk purchase discount on large orders
 - Carrying costs are proportional to average inventory
- (01 mark)**
- (v) Which of the following statements about the learning curve in labour costing is the most accurate?
- Labour hours per unit decrease by a constant absolute amount whenever cumulative production increases
 - Total labour hours are always reduced by half when output doubles
 - The average labour time per unit declines by a constant percentage each time cumulative output increases
 - The learning curve applies only under a piece-rate wage system
- (01 mark)**
- (vi) Which of the following best distinguishes safety stock from minimum stock?
- Safety stock is held for quality checks, while minimum stock is held to meet demand
 - Safety stock is extra stock for uncertainty, while minimum stock is the lowest allowable stock before reordering
 - Safety stock equals reorder level, while minimum stock equals the EOQ
 - Both are the same and are used interchangeably in inventory control
- (01 mark)**
- (vii) A company plans a product with a market price of Rs. 1,200 per unit. Details are as follows:
- | | |
|------------------------------------|--|
| ▪ Estimated demand = 200,000 units | ▪ Capital investment = Rs. 300 million |
| ▪ Management requires 12% ROI | ▪ Estimated unit cost = Rs. 1,050 |
- What % reduction in estimated unit cost is needed to meet the target cost?
- (a) 2.50% (b) 2.86% (c) 2.94% (d) 3.02%
- (02 marks)**
- (viii) After introducing Activity-Based Costing (ABC), a company finds that low-volume, customized products consume disproportionately more overhead resources compared to high-volume, standardized products. Based on this insight, which strategic response is most appropriate?
- Discontinue all low-volume customized products, as they must be unprofitable
 - Review the pricing of low-volume customized products or redesign processes to reduce their overhead drivers
 - Continue using traditional absorption costing to ensure consistency with prior costing information
 - Apply uniform cost-reduction measures across all products, regardless of their overhead consumption
- (01 mark)**

QUESTION 2

Bear Ltd (BL) is engaged in the production of 'Chem-B', which passes through two departments: Department A and Department B. Below are the production details of Department B for the latest quarter:

	Units	% of completion
Opening work-in-process	20,000	30%
Units transferred from Department A	180,000	-
Units transferred to finished goods store	160,000	-
Closing work-in-process	30,000	30%

Materials are added in Department B immediately after inspection, which occurs when the production is 35% complete. Conversion costs are incurred evenly throughout the process. 4% of the inspected units are normally lost.

Required:

Compute equivalent production units if BL uses:

- FIFO method **(04 marks)**
- Weighted average method **(04 marks)**

QUESTION 3

The following information has been extracted from the product-wise budget prepared by Empire Industries Limited (EIL) for the next year:

	Product X	Product Y
Sales/production volume (units)	120,000	160,000
	----- Rs. per unit -----	
Sales price	500	400
Cost of production:		
▪ Material	200	180
▪ Labour	100	90
▪ Factory overheads (60% variable)	60	60
	360	330
Gross profit per unit	140	70

Additional information:

- Sales are made through a national distributor who charges a commission of 5% on sales. In addition, an amount of Rs. 800,000 per month is paid to the distributor as rent for the warehouse used to store the goods. The cost of transporting goods from the factory to the warehouse is borne by EIL and is estimated at Rs. 10 per unit. All other selling costs are borne by the distributor.
- Skilled labour is used for the production of Product X and is paid a fixed monthly salary, regardless of production level. However, the labour of Product Y is semi-skilled and is paid based on production levels. Labour for Products X and Y is paid at Rs. 250 and Rs. 200 per hour, respectively.
- Total administration expenses are estimated at Rs. 600,000 per month.
- The sales ratio is expected to remain the same irrespective of any changes in total sales volume.

Required:

Compute the total sales amount at which EIL is expected to break even, along with the product-wise sales quantity. **(09 marks)**

QUESTION 4

Workhard Limited is engaged in manufacturing three products and uses Activity-Based Costing for the allocation of factory overheads. The details of its overheads for April 20X5 are estimated as follows:

Description	Rs. in '000
Salaries of labour supervisors	14,400
Fuel and power	13,320
Repairs and maintenance - machines	9,000
Repairs and maintenance - factory	6,360
Ordering department's costs	3,300
Other factory overheads	1,275
	47,655

The following data is estimated for allocating factory overheads:

	Product A	Product B	Product C
Production (units)	50,000	55,000	60,000
Raw material purchases (kg)	75,000	110,000	150,000
Order quantity (kg)	25,000	27,500	30,000
Direct labour hours per unit	1.25	1.50	1.75
Number of machines	30	40	50
Machine hours per unit	0.30	0.36	0.42
Energy consumption per machine hour (kWh)	6	10	15
Area occupied by each machine (square yards)	20	18	16

Other factory overheads are to be distributed based on production.

Required:

Compute the factory overhead rate per direct labour hour for each product using Activity-Based Costing. (08 marks)

QUESTION 5

DML produces a single product and is experiencing a decline in workers' productivity. DML currently employs 200 workers, each earning Rs. 250 per hour. On average, a worker works 25 days per month, with 8 hours per day. The following information is available in this regard:

(i) **Current operational overview for February 20X5:**▪ **Production output analysis**

	Current	Standard
Average good units per worker	720	810
Rejected units as a % of total units	10%	10%

▪ **Pricing and cost**

	Rupees
Sale price per good unit	400
Sale price per rejected unit	50
Material cost per unit	120
Variable factory overhead per labour hour	200

- Rejected units are identified only after the production process is 100% complete.

(ii) **Proposed incentive plan:**

To encourage high productivity and discourage low-quality work, management is considering implementing the following plan:

- If the workers produce above 810 units, they will be considered 'improved workers' and will be paid a bonus of Rs. 200 per unit for such extra good units.
- If the rejected unit percentage of the workers exceeds 10%, a deduction of Rs. 50 per unit will be made for such extra rejected units. The deduction applies only to workers eligible for a bonus.

The impact of the incentive plan is estimated as follows:

Category	% of workers	Average good units per worker (per month)	Rejected units as a % of total units
Improved workers	60%	880	12%
Other than improved workers	40%	630	10%

Required:

Evaluate and recommend whether adopting the incentive plan would be beneficial for DML. (08 marks)

QUESTION 6

TRM Limited uses a single raw material for production and stores it in a nearby rented warehouse. During the year 20X4, 1,600,000 units were purchased at Rs. 750 per unit, with an average order quantity of 64,000 units. TRM incurred the following costs related to ordering and holding the raw material:

	Rupees
Fixed costs of the purchase department	1,680,000
Variable costs of the purchase department	750,000
Warehouse rent	1,240,000
Transportation cost (Rs. 45,000 per order)	1,125,000
Insurance costs (2% of the purchase costs)	600,000

TRM's new finance manager believes that some of these costs can be reduced significantly if each order is based on Economic Order Quantity. In this respect, he has gathered the following information relating to 20X5:

- (i) Annual warehouse rent would comprise Rs. 300,000 plus Rs. 15 per unit for storing each unit.
- (ii) Raw material price and usage are expected to increase by 20% and 25% respectively.
- (iii) Insurance costs would remain 2% of the purchase costs.
- (iv) All other expenses are expected to increase by 10% due to inflation.

TRM maintains a safety stock of 8,000 units.

Required:

Determine the Economic Order Quantity for the year 20X5.

(07 marks)

SECTION B (Ranging from 45 to 55 marks)

QUESTION 7

Sarzameen Company Limited (SCL) uses a single raw material for its manufacturing operations. It follows the perpetual inventory system for recording and applies the weighted average method for valuing its raw material.

The following is a schedule of purchases and consumption of raw material for February 20X5:

Notes:

- (i) Insurance in transit of Rs. 91,800 was incurred on shipment received on 6 February 20X5, which was incorrectly debited to factory overheads.
- (ii) On 12 February 20X5, 1,000 kg of raw material issued for consumption exceeded the requirement. The production department returned this excess quantity to the warehouse on the same day. This returned quantity was subsequently reissued on 24 February 20X5, in addition to the issuance of 25,500 kg of raw material. No adjustment was made for the subsequent issuance of these 1,000 kg, as no entry was made on the return of these quantities.
- (iii) On 18 February 20X5, 2,000 kg of raw material were lost during transit. However, the recording in the inventory system was made based on the supplier's invoice, which listed the full quantity. SCL received Rs. 3,350,000 from the insurance company for the transit loss, which was credited to other income net of insurance-in-transit cost of Rs. 150,000. No further adjustments were made in the records to account for this shortfall.
- (iv) On 25 February 20X5, 500 kg of raw material was moved to another warehouse but was erroneously recorded as consumption.
- (v) A physical stock check of raw material at all warehouses was carried out on 28 February 20X5, which revealed the following:

Good quantity	kg	5,000
Damaged quantity	kg	400

Damaged quantity of raw material can be sold for Rs. 1,000 per kg.

SCL accounts for adjustments for inventory excess/shortage, net realisable value, and insurance shortfalls in the Profit and Loss Account.

Required:

- (a) Prepare a revised schedule for February 20X5. (11 marks)
 (b) Record necessary journal entries for notes (i) to (v). (07 marks)

QUESTION 8

Junaid Ali Limited (JAL) is engaged in the production of various products. Below are the extracts from its next year's budget relating to its products Cee and Dee:

Description	Cee	Dee
Units produced and sold	300,000	200,000
	----- Rs. in million -----	
Sales	1,200	1,000
Cost of sales		
Material	500	400
Direct labour	300	270
Factory overheads		
Indirect labour	40	30
Fuel and power	50	40
Repairs & maintenance	20	10
Other FOH (25% are fixed)	60	54
Administration expenses (Fixed)	180	150
Total expenses	1,150	954
Net profit	50	46

After preparing the budget, the price of the raw material used in Cee increased unexpectedly by 60%. The management believes that the sale price of Cee cannot be increased by more than 5% as presently the market is experiencing a surplus of Cee. Therefore, the management is considering stopping the production of Cee for about one year, after which, the production may resume as the competition is expected to subside, and a price increase would become possible.

The following information is available for making this decision:

- Cee is also used internally for the production of Dee. One unit of Cee is consumed in producing 5 units of Dee, besides other raw materials. An alternative material, available at Rs. 4,500 per unit, can be used in place of Cee. In the budget, internal transfer of Cee for the production of Dee has been recorded at the normal selling price of Cee and is included in the units sold of Cee.
- Direct labour is paid at Rs. 250 per hour, and each labour works on an average of 200 hours per month. According to the union agreement, Rs. 50,000 is payable to each labour upon termination of their services.
- 80% of indirect labour represents the salaries of labour supervisors. One labour supervisor is required for every 10 direct labours. Services of a labour supervisor can be terminated upon payment of Rs. 100,000. All remaining indirect labour is fixed.
- 90% of fuel and power is variable, as the remaining fuel and power is used for air-conditioning and other similar activities.
- 60% of repairs and maintenance pertain directly to machines. Out of the remaining amount, 30% is fixed.
- Assume that direct labour and labour supervisors required for one product cannot be utilized for another product within the company.

Required:

Based on the financial information given above, compute the change in overall net profit in each of the following situations:

- (a) If JAL continues the production of Cee (03 marks)
 (b) If JAL discontinues the production of Cee (12 marks)

QUESTION 9

Panther Limited (PL) is engaged in the production and sale of industrial chemicals. Below is the information related to one of its products, 'Panther Premium', for February 20X5:

Standard cost per batch:

Cost component	Amount (Rs.)	Rate (Rs.)
Raw material A	70,000	200 per kg
Raw material B	21,000	140 per kg
Direct labour	60,000	300 per hour
Factory overheads (25% fixed)	120% of the direct labour cost	

Actual results for February 20X5:

Cost component	Amount (Rs.)	Quantity consumed
Raw material A	26,106,000	137,400 kg
Raw material B	7,084,000	46,000 kg
Direct labour	23,000,000	-
Fixed overheads	7,800,000	-
Variable overheads	7% over standard cost per unit	

Additional information:

- (i) Each batch produces 100 units of Panther Premium.
- (ii) Direct labour rate increased by 5% due to inflation.
- (iii) PL produced 37,500 units of Panther Premium, which were 1,500 units over its budgeted quantities.

Required:

Calculate the following:

- (a) Material price, mix, and yield variances (06 marks)
- (b) Direct labour rate and efficiency variances (03 marks)
- (c) Variable overhead expenditure and efficiency variances (03 marks)
- (d) Fixed overhead expenditure, efficiency, and capacity variances (05 marks)

(THE END)